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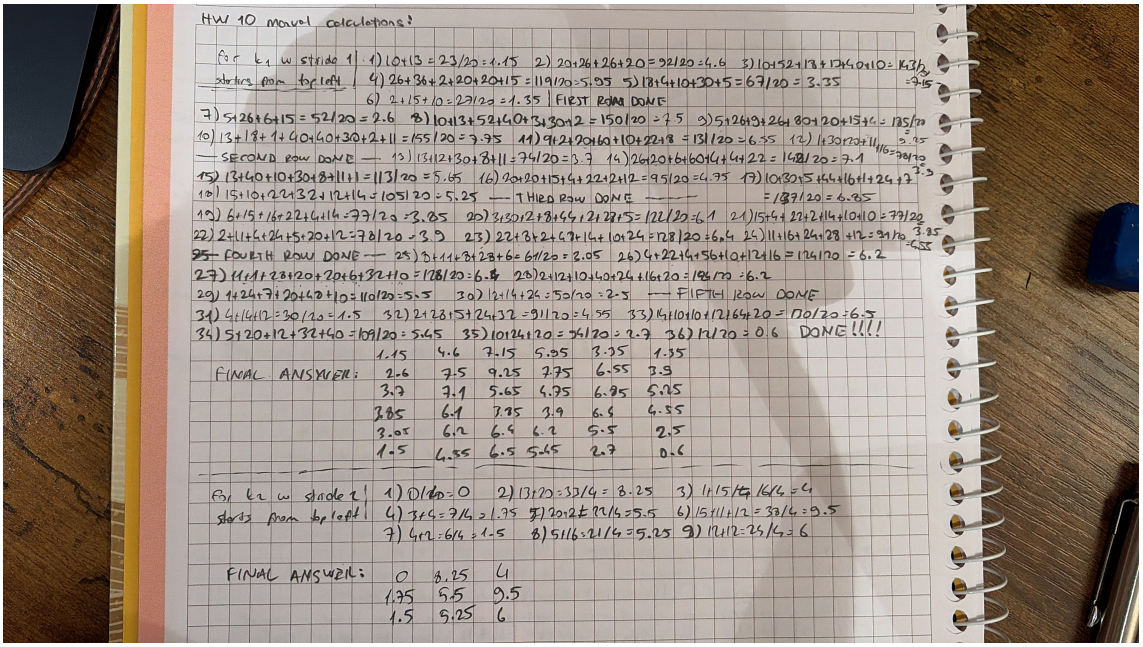
Assignment Sheet 10

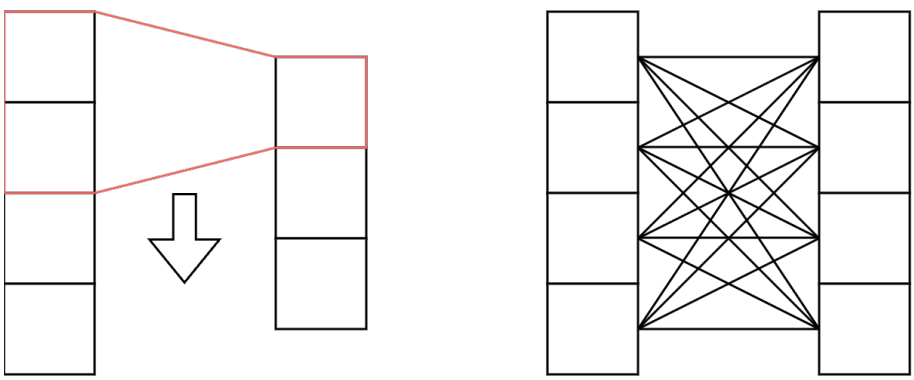
Question 1: Convolution by Hand (6 points)

(a) Convolve the two kernels by hand



Question 2: 1-D Conv vs. FC Layer (8 points)

(a)



1D convolution fully connected

(b) $\frac{\partial \mathcal{L}}{\partial K_j}$ (Conv layer)

We apply the chain rule. Since the weight K_j is shared across all spatial positions i, we sum the gradients from all outputs Z_i that use K_j.

$$\frac{\partial \mathcal{L}}{\partial K_j} = \sum_i \frac{\partial \mathcal{L}}{\partial Z_i} \frac{\partial Z_i}{\partial K_j}$$

Given $Z_i = \sum_k V_{i+k} K_k$, the derivative is $\frac{\partial Z_i}{\partial K_j} = V_{i+j}$.

$$\boxed{\frac{\partial \mathcal{L}}{\partial K_j} = \sum_i \frac{\partial \mathcal{L}}{\partial Z_i} V_{i+j}}$$

(c) $\frac{\partial \mathcal{L}}{\partial W_{j,i}}$ (Fully Connected)

We use the chain rule again. In a fully connected layer, a specific weight $W_{j,i}$ only connects input j to output i. There is no sharing.

$$\frac{\partial \mathcal{L}}{\partial W_{j,i}} = \frac{\partial \mathcal{L}}{\partial Z_i} \frac{\partial Z_i}{\partial W_{j,i}}$$

Given $Z_i = \sum_k V_k W_{k,i}$, the derivative is $\frac{\partial Z_i}{\partial W_{j,i}} = V_j$.

$$\boxed{\frac{\partial \mathcal{L}}{\partial W_{j,i}} = \frac{\partial \mathcal{L}}{\partial Z_i} V_j}$$

(d)

- **Convolutional Layer:** The gradient includes a summation $\sum_i$. This represents convolutions strategy of weight sharing, a single parameter K_j affects the whole output map, so we accumulate the error signals from all locations.
- **Fully Connected Layer:** The gradient has no summation. Each weight $W_{j,i}$ is unique to one specific connection, so its update depends only on that specific part.

Question 3: Number of parameters in CNNs (6 points)

(a)

We calculate spatial dimensions using $O = \lfloor \frac{I-K}{S} \rfloor + 1$.

- **1:** Input 24, $K = 4$, $S = 1$.

$$\frac{24-4}{1} + 1 = 21 \quad \Rightarrow \quad \mathbf{21 \times 21}$$

- **2:** Input 21, $K = 3$, $S = 2$.

$$\lfloor \frac{21-3}{2} \rfloor + 1 = 10 \quad \Rightarrow \quad \mathbf{10 \times 10}$$

- **Output:** The fully connected layer maps to **10** classes.

(b)

We count the parameters excluding bias.

- $C^{(1)}$**:** 4×4 (kernel) $\times$ 1 (input depth) $\times$ 1 (output depth) $=$ **16**.
- $C^{(2)}$**:** 3×3 (kernel) $\times$ 1 (input depth) $\times$ 1 (output depth) $=$ **9**.
- **FC Layer:** 100 (flattened input) $\times$ 10 (output) $=$ **1000**.
- **Total:** $16 + 9 + 1000 =$ **1025**.

(c)

We count every application of a weight.

- $C^{(1)}$**:** The 16 weights are applied to each pixel of the 21×21 output.

$$16 \times 441 = \mathbf{7,056}$$

- $C^{(2)}$**:** The 9 weights are applied to each pixel of the 10×10 output.

$$9 \times 100 = \mathbf{900}$$

- **FC Layer:** Each weight is applied once.

$$\mathbf{1,000}$$